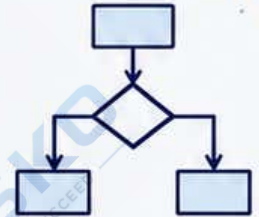


1010101010
0101010101
1010101010
0101010101



10110
01001
10101



Most Important Questions

INSTRUCTIONS



1. Understand the Concepts

Go through the basic concepts related to each question before practising.



2. Practice Regularly

Solve these important questions and their variations for better understanding.



3. Focus on Exam Pattern

These questions are selected as per previous year papers and latest syllabus.



4. Attempt Within Time Limit

Practice under timed conditions to improve speed and accuracy.



5. Revise and Self-Check

Review your answers, identify weak areas and revise regularly to score higher.

Learn • Practice • Improve • Succeed

YOUR HARD WORK PAYS OFF!



Programming with C

Unit I - Basics of C

- **Answers in one or two lines (1 mark questions)**

1. What is an algorithm?
2. What is a flowchart?
3. Name the types of programming languages.
4. Who developed the C language?
5. State any two advantages of C.
6. What is a keyword?
7. What is an identifier?
8. What is a constant?
9. What is a variable?
10. What is type conversion?

- **Short Answer Questions (3 marks questions)**

1. Explain algorithm and flowchart with suitable examples.
2. Explain the structure of a C program.
3. Explain character set, identifiers and keywords.
4. Explain constants and variables.
5. Explain qualifiers and type conversion.
6. Explain operators and expressions in C.

- **Long Answer Questions (6 marks questions)**

1. Explain the structure of a C program with suitable example.
2. Explain different types of programming languages and the history of C language.
3. Explain operators and expressions in C with examples.
4. Explain constants, variables, symbolic constants, qualifiers and type conversion.

Unit II - I/O, Branching & Looping

- **Answers in one or two lines (1 mark questions)**

1. What is formatted I/O?
2. What is `scanf()`?
3. What is `printf()`?
4. What is unformatted I/O?
5. What is `getch()`?
6. What is `gets()`?
7. What is the conditional operator?
8. What is `switch` statement?
9. What is a loop?
10. What is the `break` statement?
11. What is the `continue` statement?
12. What is the `goto` statement?

- **Short Answer Questions (3 marks questions)**

1. Explain `scanf()` and `printf()` with examples.
2. Explain `getch()`, `getchar()`, `gets()`, `putch()`, `putchar()` and `puts()`.
3. Explain if-else and nested if statements.
4. Explain conditional operator (`?:`).
5. Explain `switch` statement with examples.
6. Explain `while`, `do-while` and `for` loops.
7. Explain `break`, `continue` and `goto`.

- **Long Answer Questions (6 marks questions)**

1. Explain branching statements in C with suitable examples.
2. Explain looping statements (`while`, `do-while`, `for`) with examples.
3. Explain formatted and unformatted I/O functions in C.
4. Explain `if`, `if-else`, nested-if, conditional operator and switch statement.

5. Explain loop control statements and nested loops.

Unit III - Arrays, Structure & Pointers

- **Answers in one or two lines (1 mark questions)**

1. What is an array?
2. What is a one-dimensional array?
3. What is a two-dimensional array?
4. What is a structure?
5. What is a union?
6. What is a nested structure?
7. What is a pointer?
8. What is pointer arithmetic?
9. How is an array declared in C?
10. How is a pointer initialized?

- **Short Answer Questions (3 marks questions)**

1. Explain declaration and initialization of one-dimensional arrays.
2. Explain two-dimensional arrays with example.
3. Explain structure declaration, initialization and definition.
4. Explain nested structures.
5. Differentiate between structure and union.
6. Explain pointer declaration and initialization.
7. Explain pointer arithmetic.

- **Long Answer Questions (6 marks questions)**

1. What is an Array ? Explain one-dimensional and two-dimensional arrays with declaration and initialization.
2. Explain structure in C with declaration, initialization and suitable example.
3. Explain pointers in C with declaration, initialization and pointer arithmetic.

4. Explain nested structures and union with examples.
5. Differentiate between structure and union in detail.

Unit IV - Operating System

● Answers in one or two lines (1 mark questions)

1. What is a function?
2. What is a function prototype?
3. What is an actual argument?
4. What is a formal argument?
5. What is call by value?
6. What is call by reference?
7. What is a local variable?
8. What is a global variable?
9. What is recursion?
10. What is a string?
11. What is `strlen()`?
12. What is `strcpy()`?
13. What is `strcmp()`?
14. What is `strcat()`?

● Short Answer Questions (3 marks questions)

1. Explain function prototype and function calling.
2. Explain actual and formal arguments.
3. Differentiate between call by value and call by reference.
4. Explain categories of functions in C.
5. Explain local and global variables.
6. Explain functions with arrays.
7. Explain function recursion with example.
8. Explain `strlen()`, `strcpy()`, `strcmp()` and `strcat()`.

- **Long Answer Questions (6 marks questions)**

1. Explain functions in C, including function prototype, arguments and function calling.
2. Explain call by value and call by reference with suitable examples.
3. Explain different categories of functions in C with examples.
4. Explain recursion and functions with arrays in C.
5. Explain string functions `strlen()`, `strcpy()`, `strcmp()` and `strcat()` with examples.

FINAL MUST-PREPARE LIST

UNIT I

- ★ Structure of C Program
- ★ Operators & Expressions
- ★ Algorithm & Flowchart
- ★ History + Advantages of C
- ★ Constants, Variables & Type Conversion

UNIT II

- ★ Formatted & Unformatted I/O
- ★ Branching Statements
- ★ Looping Statements
- ★ `switch`
- ★ `break`, `continue`, `goto`
- ★ Nested Loops

UNIT III

- ★ 1D & 2D Arrays
- ★ Structure
- ★ Structure vs Union
- ★ Pointers
- ★ Pointer Arithmetic
- ★ Nested Structure

UNIT IV

- ★ Functions & Function Prototype
- ★ Call by Value vs Call by Reference
- ★ Categories of Functions
- ★ Recursion
- ★ Functions with Arrays
- ★ String Functions (`strlen`, `strcpy`, `strcmp`, `strcat`)